Journal of Ethnopharmacology ∎ (∎∎∎∎) ∎∎∎–∎∎∎

Contents lists available at ScienceDirect

Journal of Ethnopharmacology

journal homepage: www.elsevier.com/locate/jep

Development of Ayurveda – Tradition to trend

Pulok K. Mukherjee [a,*], Ranjit K. Harwansh [a], Shiv Bahadur [a], Subhadip Banerjee [a], Amit Kar [a], Joydeb Chanda [a], Sayan Biswas [a], Sk. Milan Ahmmed [a], C.K. Katiyar [b]

[a] School of Natural Product Studies, Department of Pharmaceutical Technology, Jadavpur University, Kolkata 700032, India
[b] Research & Development Center, Healthcare Division, Emami Limited, 13, BT Road, Kolkata 700056, India

ARTICLE INFO

Article history:
Received 15 February 2016
Received in revised form
26 July 2016
Accepted 11 September 2016

Keywords:
Ayurveda
Chemo-profiling
Ayurvedic drug
Quality evaluation
Evidence based validation

ABSTRACT

Ethnopharmacological relevance: Ayurveda entails a scientific tradition of harmonious living and its origin can be traced from ancient knowledge in *Rigveda* and *Atharvaveda*. Ayurveda is a traditional healthcare system of Indian medicine since ancient times. Several Ayurvedic medicines have been exploiting for treatment and management of various diseases in human beings. The several drugs have been developed and practiced from Ayurveda since ancient time to modern practice as 'tradition to trend'. The potential of Ayurvedic medicine needs to be explored further with modern scientific validation approaches for better therapeutic leads.
Aim of the study: The present study was aimed to explore the various aspects of Ayurveda and inspired drug discovery approaches for its promotion and development.
Materials and methods: We have reviewed all the literature related to the history and application of Ayurvedic herbs. Various aspects for the quality control, standardization, chemo-profiling, and metabolite fingerprinting for quality evaluation of Ayurvedic drugs. The development of Ayurvedic drugs is gaining momentum with the perspectives of safety, efficacy and quality for promotion and management of human health. Scientific documentation, process validation and several others significant parameters are key points, which can ensure the quality, safety and effectiveness of Ayurvedic drugs.
Results: The present review highlights on the major goal of Ayurveda and their significant role in healthcare system. Ayurveda deals with several classical formulations including arka, asavas, aristas, churna, taila, vati, gutika, bhasma etc. There are several lead molecules that have been developed from the Ayurvedic herbs, which have various significant therapeutic activities. Chemo-profiling of Ayurvedic drug is essential in order to assess the quality of products. It deals with bioactive compound quantification, spurious and allied drug determination, chromatographic fingerprinting, standardization, stability and quality consistency of Ayurvedic products.
Conclusions: Scientific validation and the documentation of Ayurvedic drugs are very essential for its quality evaluation and global acceptance. Therapeutic efficacy of Ayurvedic herbs may be enhanced with high quality, which can be achieved by identity, purity, safety, drug content, physical and biological properties. Ayurvedic medicines need be explored with the modern scientific approaches for its validation. Therefore, an attempt has been made in the present review to highlight the crucial aspects that need to be considered for the promotion and development of Ayurvedic medicine.

1. Ayurvedic system of medicine and its role in healthcare

India has a rich heritage of traditional system of medicine. Ayurveda is the traditional Indian system of medicine which is meant not only for curing the diseases but also for prevention of the occurrence of illnesses. Ayurveda provides a plethora of information on ethnic folklore practices and traditional aspects of therapeutically important medicines. Ayurveda is getting global acceptance primarily due to its holistic therapeutic practice, extensive profound conceptual basis and survival of its medicines since prehistoric times. This concept of drugs and formulations developed in ancient times still finds its relevance in spite of changes in the environment, lifestyle, culture and disease patterns (Mukherjee and Wahile, 2006; Mukherjee et al., 2016). The basic principle of Ayurvedic treatment comprises of two essential parts. These are to prevent the cause of disease and to make the patient more aware about the cause of the disease. The main objective of Ayurveda has been explained in Fig. 1.

Ayurveda treats a patient as a whole and not the disease alone. This system of medicine emphasizes the uniqueness of each

* Corresponding author.
E-mail address: naturalproductm@gmail.com (P.K. Mukherjee).

http://dx.doi.org/10.1016/j.jep.2016.09.024

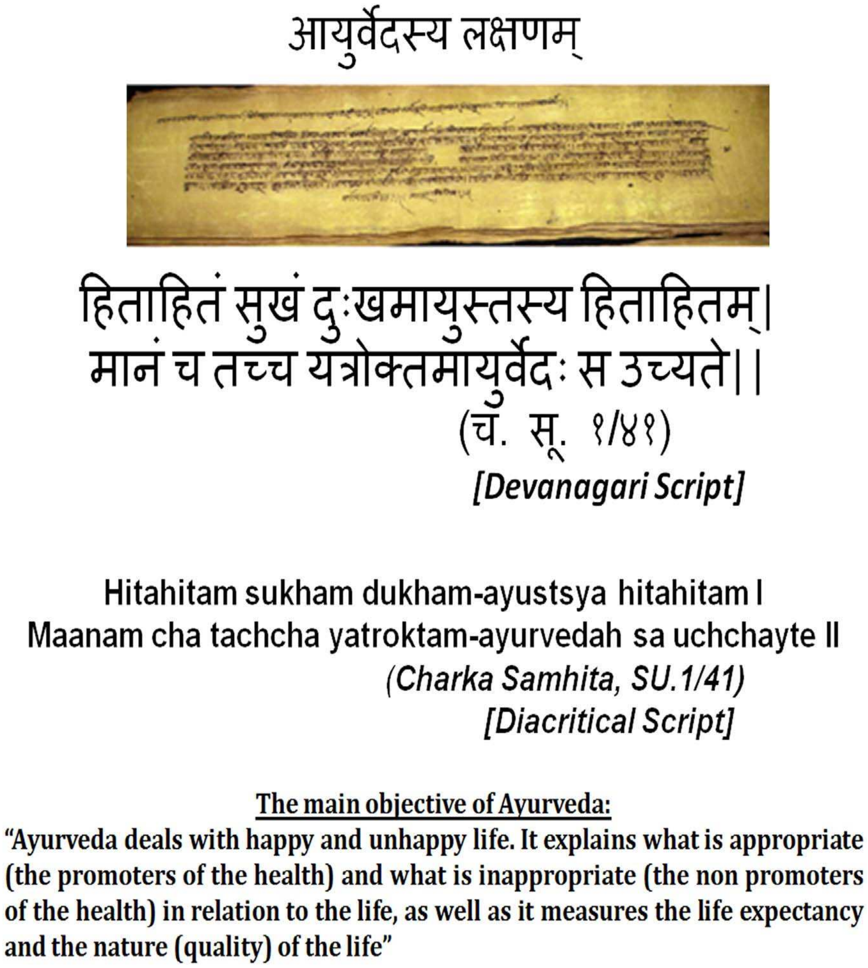


Fig. 1. Main goal of Ayurveda.

person with regard to bio-identity, socio-economical status, biochemical and physiological conditions, which may lead to a particular type of illness. Ayurveda needs further exploration with modern scientific approaches for addressing various healthcare issues. Ayurvedic system adopts a holistic approach towards healthcare by balancing the physical, mental and spiritual functions of the human body. Science of Ayurveda is unique as it provides opportunity to make a healthy, harmonious and long life (Krishnamachary et al., 2012). Over the past few decades, research on Ayurveda through various endeavours has given rise to interdisciplinary research programmes employing several disciplines in this field. Several lead molecules, ready-to-use products and processes are emerging. It is quite popular among people due to their practical benefits, traditional beliefs, economical advantage and easy access (Mukherjee and Houghton, 2009).

The development of Ayurvedic medicine (AM) is gaining momentum in keeping with the perspectives of safety, stability, efficacy and quality for betterment of human health. Medicinal plants serve as most valuable source for remedy of many diseases. The increasing search for therapeutic agents derived from plant species is justified by the emergence of new diseases. Bioactive compounds from different medicinal plants have a major role in the management and improvement of human health (Mukherjee et al., 2012).

Ayurvedic medicines are used as raw, crude materials, extracts and preparations for therapeutic purposes. Authentication, quality control, standardization, chemo-profiling, process validation, regulatory aspects, clinical risk assessment, consumer awareness and post marketing surveillance are the key points which could ensure the quality, safety and effectiveness of Ayurvedic medicines (Mukherjee et al., 2014, 2015). Therefore, the present article highlights on the several aspects of validation and quality evaluation of Ayurvedic medicine for its worldwide acceptance.

1.1. Objectives of Ayurveda

प्रयोजनंचा स्यस्वस्थस्यस्वा स्थ्यरक्षणमातुरस्यवकिारप्रशमनंच॥२६॥.

Ch. Su. – 30/26.

[The object (of Ayurveda) is to protect health of the healthy.

and to alleviate disorders in the diseased].

The basic objective of Ayurveda as mentioned by Charaka is dual in nature. In the ancient script *Charaka Samhita* (*Sutra Sthana* 30/26) (as described in above *shloka*) it is enumerated that the principle objective of Ayurveda is to protect the health of the healthy and to alleviate the disorders in the diseased condition (Sharma, 1992).

1.2. Definition of health according to Ayurveda

शरीरेन्द्रयिसत्त्वात्मसंयोगो धारि जीवतिम्।.

नत्यिगश्चानुबन्धश्च पर्यायैरायुरुच्यते॥.

Ch. Su. – 1/42.

['*Ayus*' means the conjunction of body, sense organs, mind and self and is known by the synonyms *dhari, jivita, nityaga and annubandha*. Ayurveda defines life as the combination of *Shareera* (body), *Indriya* (sense organs), *Satwa* (mind) and *Atma* (soul)].

Ayurveda is a complete medical system which deals with all aspects of physical health, mental balance, spiritual well-being, social welfare, environmental considerations. It also focuses on dietary and lifestyle habits which are controlled by daily living trends and seasonal variations. Ayurveda describes about treating and managing specific diseases while appreciating the nature, life and the empowerment of the individual towards sustainable living by following a holistic approach. The Ayurvedic concept of health and disease considers the *Tridosha* theory as conscientious for the biological entity in a balanced state. The structural, functional and behavioral dimensions of the living organism are being conceptualized with *Doshas* (humoral component) viz., *Sharirika* (somatic) - *Vata, Pitta, Kapha* and *Manasika* (psychic) - *Satva, Rajas, Tamas. Sharirika* and *Manasika doshas* for the maintenance of physiological homeostasis are interrelated to each other on psychosomatic dimensions. In Ayurveda, the three "*Doshas*" (*Vata, Pitta* and *Kapha*) give a rationalization for all the psychobiological functions of living beings. They are called "*doshas*" (i.e., "defective" or "deranged" elements) because they are well organized elements that are highly susceptible to imbalance and derangement. Diverse qualities and functions have been attributed to each *dosha*. The kinetic mechanism of a system has been attributed to *Vata*, the metabolic mechanism to *Pitta*, the structural and stability mechanism to *Kapha*. For instance, *Vata* contributes to the expression of shape, cell division, signaling, movement and excretion of wastes.

1.3. *Major treatment of classification in Ayurveda*

तस्यायुर्वेदस्याङ्गान्यष्टौ; तद्यथा- कायचकित्िसाशालाक्यं,शल्यापहर्तृकं,

वषिगरवैरोधकिप्रशमनं, भूतवद्िया, कौमारभृत्यकं, रसायनं, वाजीकरणमतिि॥.

Ch. Su. – 30/28.

[*Ayurveda has eight divisions such as – Kayacikitsa (medicine), Salakya (dealing with diseases of supra-clavicular region), Salyapahartrka (dealing with extraction of foreign bodies), Visagara-Vairodhika-Prasamana (dealing with alleviation of poisons, artificial poisons and toxic symptoms due to intake of antagonistic substances), Bhutavidya (dealing with spirits or organisms), Kaumarabhrtya (Pediatrics), Rasayana (dealing with promotive measures) and Vajikarana (dealing with aphrodisiacs)*].

The philosophy of Ayurveda in clinical practice is known as *Ashtanga Ayurveda*. Major disciplines in Ayurveda are such as *Ayurveda Siddhanta* (fundamental principles of Ayurveda), *Ayurveda Samhita* (dealing with Ayurvedic classics), *Sharira Rachana* (anatomy), *Sharira Kriya* (physiology), *Dravyaguna Vigyan* (Materia Medica and pharmacology), *Rasashastra* (metal and minerals processing), *Bhaishajya Kalpana* (pharmaceuticals), *Kaumarabhritya* (pediatrics), *Prasutitantra* (obstetrics and gynecology), *Swasthavritta* (social and preventive medicine), *Kayachikitsa* (internal medicines), *Roganidana* (etiopathology), *Shalyatantra* (surgery), *Shalkyatantra* (eye and ENT), *Manasaroga* (psychiatry), *Agadatantra* (toxicology and forensic medicine), *Sangaharana* (anesthesia), and *Panchakarma* (cleansing for rejuvenation therapy) (Mukherjee et al., 2016). The major discipline in Ayurveda has been explained in Fig. 2.

Four super specialties such as *Bhutavidya* (demonology), *Sarpavidya* (dealing with poisons and venoms), *Rasayana* (rejuvenation) and *Vajikarana* (dealing with aphrodisiacs) existed in Vedic literature prior to written systemic development of Ayurveda (Kutumbiah, 1962). After establishment of *Atreya Sampradaya* (school of physician) and *Dhanwantari Sampradaya* (school of surgeons) four more disciplines were included on the development of reason based Ayurveda like *Kayachikitsa* (internal medicine), *Shalyatantra* (surgical sciences), *Shalakyatantra* [eye and ear nose, and throat (ENT)] and *Kaumarabhritya* (maternal and child health). *Sarpavidya* was restructured and renamed as *Agadatantra* (toxicology and forensic medicine) (Misra, 2007; Srikanthamurthy, 2010).

There are three basic principles of Ayurveda i.e. *Roga vigyan, Vikriti vigyan* (science of disease process) and *Chikitsa vigyan* (therapeutic modalities). The pathological processes are described as *Panchanidana* (five etiological factors): these are *Nidana* (cause), *Purvarupa* (premodial symptoms), *Rupa* (symptomatology), *Upasaya* (therapeutic measures) and *Samprapti* (pathogenesis) (Roy and Debnath, 2000; Srikantamurthy, 2003). The main aim of *Chikitsa* (therapeutic management/treatment) is to convert unhappiness to happiness fundamentally guided by *Samanya* (general) and *Visesha* (particular) principle (Singh, 2005).

2. History of Ayurveda

2.1. *Mythological and Vedic history*

The genesis of Ayurveda is ancient, coupled with the pursuit of good health which is as old as human existence, fighting to live and adapt to nature. It evolved with the development of human civilization, hence emerged as a comprehensive system of healthcare. The important events of the history of Ayurveda are summarized in Fig. 3. Indian philosophy articulates health as the prerequisite for attaining materialistic, spiritual and social improvement of the human being. Brahma (Hindu mythological God of Creation) who is believed to be the creator of the universe evolved the science of Ayurveda by meditation and taught it to Prajapati Daksha. The Vedas including *Rig Veda, Sam Veda, Yajur Veda and Atharva Veda* are the oldest (5000–1000 BCE) Indian literatures which have references regarding plants and natural resources for various treatment. In the ancient literature of *Ramayana* and *Mahabharata* different feats of medicine using plants and surgical procedures are observed. But Ayurveda got established during the "Samhita" period as a full-fledged medical system during 1000 BCE. *Charaka* and *Sushruta Samhita* were written during all these period.

The origin of the two *Samhitas* are connected to the two different school of thoughts which were developed by two prominent instructors - sage Bharadwaja and sage king Devodas Dhanwantari. Sage Bharadwaja taught Ayurveda to Punarvasu Atreya, who later formed the Atreya School or the School of Physicians. The latter, the ascetic king Devodas Dhanwantari (who is said to have been the incarnation of the physician-God Dhanwantari) formed the Dhanwantari School or the School of Surgeons. Of these two schools, the great exponents of the school of physicians were six disciples of sage Atreya-Agnivesha, Bhela, Jatukarna, Parasara, Harita and Ksharpani. Each of them wrote a comprehensive work known after his name on the Ayurvedic practice of medicine. Agnivesha, the best-known disciple of Punarvasu Atreya who wrote the Agnivesha-Samhita before 6th BCE. This book was received by Charaka in 1st CE who is said to have been the trusted physician of the king Kanishka. However, he could not complete his task and left it at a point in the chapter "Chikitsasthana". Later in 8th Century CE a scholar belonging to Kashmir, Dridhabala, added 17 chapters called "Siddhisthana" and "Kalpasthana" to complete Charaka's book- the 'Charaka Samhita,' which is now considered as the most ancient and authoritative text in Ayurveda.

The exponents of the school of surgeons were the disciple of Dhanwantari, The king of Varanasi, the main amongst others were, Susruta, Bhoja, Gopura, Karavirya, Aupadhenava, Aurabhra,

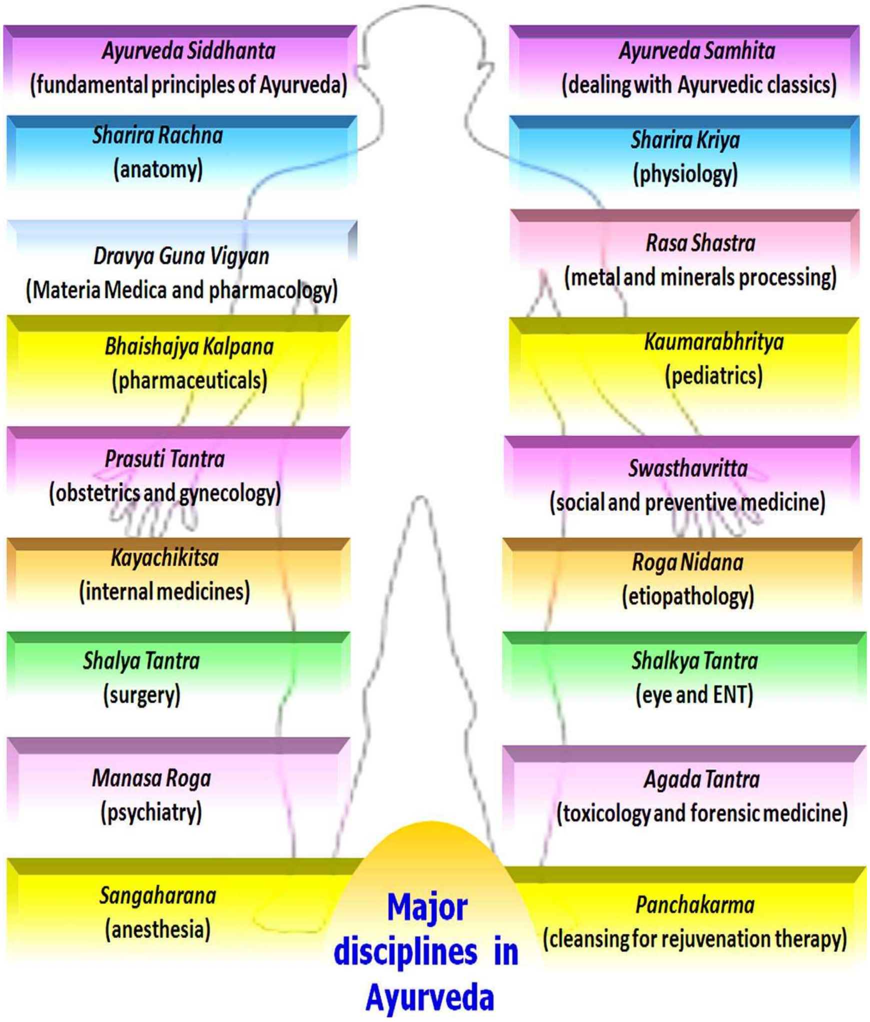


Fig. 2. The major disciplines in Ayurveda.

Vaitaran, Rakshita, etc. Each of them wrote a comprehensive work on the practice of surgery and midwifery, some of these works still occur extensively in later works. Like the *Charaka Samhita* and *Sushruta Samhita* were also written around the same period before the 6th century CE. In contrast with *Agnivesha Samhita* which is primarily a book on general medicine, Sushruta's work deals mainly with surgical matters, where we get detailed information about surgical instruments of ancient India. Later, some anonymous writer added the *Uttara-tantra*. In the present day the complete work, inclusive of the supplement is known as the *Sushruta Samhita* (Anonymous, 2012). The essentials of *Charaka* and *Sushruta Samhita* were assembled in the treatises of *Astanga Sangraha* and *Astanga Hridaya*, which was written by Vrdha Vagbhata and Vagbhata in the 6th–7th Century CE. Thus, the three top books stated above are cumulatively known as the "Brihattrayi" formed the foundation for subsequent classics like *Madhava Nidana*, *Sarangadhara Samhita*, and *Bhavaprakasha*, which are known as "Laghutrayi." *Madhava Nidana* was written by Madhavakara deals with the diagnosis of diseases. *Sarangadhara Samhita* enumerates pharmaceutics of Ayurveda focused on dosage forms and formulations. *Bhavaprakash* written by Bhava Misra highlighted on medicinal plants and diets.

During Buddhist period, a famous surgeon known as Jivaka who used to treat Gautam Budha studied Ayurveda at Takshashila University. It was around 200 BCE medical students from different parts of the world used to come to the ancient University of Takshashila to learn Ayurveda. From 200 to 700 CE, University of Nalanda attracted medical students from China and Japan. It is presumed that various medical systems like Tibetian and Chinese traditional medicines of the world have been nurtured by Ayurveda which reached them through Buddhism. Greeks and Romans also came in contact with sea-trade relations. Medicinal use of metals and minerals were started by Nagarjuna around 800 CE which is known as *Rasasashtra*. Some classical treatises of *Rasasashtra* are *Rasaratnasamuchaya, Rasarnava* and *Rasa Hridaya Tantra* elaborating manufacture, detoxification of metals using various natural products and preparation of metallic formulations.

2.2. Medieval and modern history of Ayurveda

Ayurveda flourished well under the patronage of the Hindu rulers up to the advent of the Mughals in the 11th century (Jaggi, 1973). There was a succession of foreign rulers until the arrival of the British rule. By the end of the 19th century, Ayurveda was profoundly influenced by their encounters with Western Medicine (Bastos, 2009; Weiss, 2009). The end of the 19th century was marked by a rise in many books on Ayurveda in English, Sanskrit and vernacular languages (Panikkar, 2002). This outpouring of Ayurveda publications roughly coincided with the rise of nationalism in general and that of Indian revivalist nationalism in particular (Sarkar, 2001). However despite all these adversaries Ayurveda sustained its popularity among the Indian citizens and treatises were continued to be written. These compositions got boost under the patronage of the surviving Hindu stately

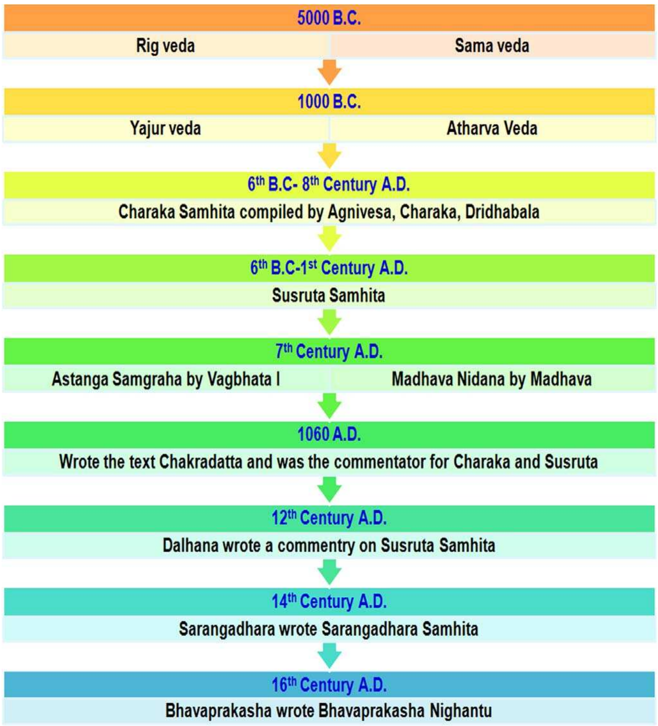


Fig. 3. Historical background of Ayurveda.

kingdoms. That is why probably we find that even today Kerala, Gujarat and West Bengal have got flourishing Ayurvedic Institutions. The contribution of Bengal in Ayurveda has been enormous. A luminary of Ayurveda from Bengal Kaviraj Ganga Prasad Sen, Mahamahopadhyaya Bijoyratna Sen, Kaviraj Amritalal Gupta and Kaviraj Kailash Chandra Sen contributed Ayurveda with books and published magazines (Srikanthamurthy, 1968). During the later part of the 19th century, it has been observed that the Indian writers have tried to interpret Ayurvedic sciences in the light of the Western medical theories and also translated the Sanskrit medical works in the regional languages. The foremost example is the Hindu Chemistry of Acharya Prafullachandra Roy, who explained ancient knowledge of Indian chemistry and also the art of pharmaceutics in Ayurveda (Ray, 1956; Gangadharan, 1982).

3. Quality evaluation of Ayurvedic drug

Quality control and standardization of raw materials and Ayurvedic drugs are the major criteria for their acceptability in modern medicine. These ensure the consumer's right to get pure, safe, potent and effective medication. The bioactive extract should be standardized on the basis of active principle or major compound(s) with chromatographic and spectroscopic techniques. The next important step is to make a stable Ayurvedic product with maximum shelf-life (Mukherjee, 2005). However there are some major aspects which need to be focused for the drug development and acceptance of Ayurveda globally in human healthcare.

According to Ayurveda, collection of plant material is a very crucial aspect for the quality of the medicinal plant because there are several factors which are needed to be considered before the collection. The specific place, part, method and time for collection are more significant to obtain best qualities of medicine. The important factors affecting the quality of drugs such as *Guna*, *Desha*, *Kala*, *Pakva-apakva avastha*, *nav-purana avastha*, *Prayojyanga*, *Karma* and *Disha* should be considered while collecting the plants materials (Tavhare and Nishteswar, 2014). The phytoconstituents of medicinal plants are available through properly executed harvesting techniques therefore it is mandatory to collect the herbs to get optimum qualities and good therapeutic effects (Chandra et al., 2014).

There are four methods of plant collection which have been described in Ayurveda as *Bhumi pariksha* (land selection), *Sangrahaniya dravyas* (drug selection), *Sangrahaniya Vidhi* (cultivating method) and *Sangrahaniya Kala* (collection time). Charaka describes the importance of *Ritu* (seasons) in germination and growth of Ayurvedic herbs. Plants should have the specific characteristic (*Rasa* in abundance i.e. Veerya, and *Gandha* for its collection in suitable seasons. Further Charaka also highlights on the consequence of stars, planets, moon, sun, air and fire on manifestation of *Rasa*, *Veerya*, *Vipaka* and *Prabhava* of drugs.

As per modern science, collection period also affects the drug potency through changes in several factors. These factor includes climate, temperature, rain fall, duration of day light, altitude, methods of cultivation, effect of lunar cycle, collection from wild area, soil condition and methods of collection, processing and storage have significant impact on the secondary metabolites (Tavhare and Nishteswar, 2014). The appropriate collection technique aims to get the genuine drug with highest purity which can improve the quality and therapeutic efficacy of Ayurvedic drugs.

3.1. Quality control of Ayurvedic medicine

Quality control of Ayurvedic medicine requires knowing what is in chemical constituents of the plant, what happens during its processing, chemical analysis and biological evaluation till the finished product reaches the consumers (Cordell, 2011). There are serious concerns being expressed all over the world with respect to contamination and adulteration of traditional medicines. Contaminants may include various pesticides, herbicides, insecticides, heavy metals (arsenic, cadmium, lead and mercury), microbial species and radiation. Adulterants may include other cheaper plant materials with similar biological effects or synthetic drugs (Mukherjee et al., 2013). All the adequate information about the botanical derived products should be considered with appropriate deliberation for analytical and batch to batch variability in the medicinal preparation. Identification, analytical specification of Ayurvedic formulation should be evaluated based on the different parameters as mentioned in the Ayurvedic Pharmacopoeia (Lavekar, 2006). The different important parameters for quality evaluation of Ayurvedic medicine are given in Table 1. Their scientific validation and documentation should be based on the various modern approaches and hyphenated technologies.

3.2. Methods of quality evaluation

Botanicals are standardized based on the presence of a known active ingredient or specific markers when the active markers are not yet recognized. But this can help in establishing the product's quality depending on the characteristic fingerprints. Plants contain several active substances in certain ratios in standardized extracts. This ratio must be kept constant, within narrow limits, from one preparation to another. The unique processing methods followed for the manufacturing of Ayurvedic products turn herbal ingredients into very complex mixtures, through which the separation, identification and estimation of chemical components become more challenging in some cases. Moreover, herbs are known to contain several components and in many cases the absolute compound responsible for the pharmacological activity is unknown (Mukherjee et al., 2016).

Standardization approaches should be taken into account for all the aspects that contribute to the quality of the Ayurvedic medicine, including correct identification of sample, pharmacognostic

Table 1
Quality control parameters of Ayurvedic drugs.

	Modern aspects	Ayurvedic aspects
Quality evaluation of Ayurvedic medicine	Description	*Darshana pareeksha*
	Colour	*Darshana pareeksha*
	Odour	*Ghrana pareeksha*
	Taste	*Rasana pareeksha*
	Loss on drying at 105 °C, total ash acid, insoluble ash, total solid, pH, volatile oils	*Bahya/rasayanika parikasha*
	Particle size, bulk density, tap density	*Darshana* and *Sparshana pareeksha*
	Heavy/toxic metal analysis	*Bahya/rasayanika parikasha*
	• Lead, cadmium, mercury, arsenic	
	Microbial analysis	
	• Total viable aerobic count,	
	• Total enterobacteriaeceae,	
	• Total fungal count	
	Test for specific pathogen	*Krimi,/desha pariksha, Panchgyanendriya pariksha*
	• *E coli*	
	• *Salmonella* sp.	
	• *S. aureus*	
	• *Pseudomonas aeruginosa*	
	Pesticide residue analysis	*Desha pariksha*
	• Organo-chlorine pesticide	
	• Organoposphorous pesticide	
	• Synthetic pyrethroids	
	Test for Aflotoxins	*Prabhava karakas*
	• (B1+B2+G1+G2)	
	TLC/HPLC/HPTLC-Profile with marker	*Darshana pareeksha*
	Tablets/Capsules	*Mana pariksha*
	• Uniformiy of weight/content	
	Disintegration time	*Darshana pareeksha*
	Friability (If tablet)	*Darshana pareeksha*
	Hardness	*Darshana pareeksha*
	Lethal dose	*Vishakta matra*
	Optimum effective dose	*Prabhava sheela matra*
	Shelf life	*Saviryata avadhi*
	Preservative	*Prakshipta dravyas*
	Active compound	*Virya*
	Binders	*Sandhana karakas*

evaluation, organoleptic evaluation, volatile matter, quantitative evaluation (ash values, extractive value, foreign matter etc.), phytochemical evaluation, xenobiotics testing, toxicity testing, microbial load testing and biological activity (Mukherjee et al., 2003). Among these phytochemical profile has special significance because it has direct effect on the activity of the herbal medicine. Quantification of marker compounds would serve as an important parameter in assessing the quality of the botanicals (Folashade et al., 2012). Different chromatographic techniques are most frequently used for the identification and quality control of herbal medicines or products. While there are multitudes of chromatographic techniques to achieve separation, the common thread is the separation of compounds through the use of variations in mobile and stationary phases. Chromatographic technique includes thin layer chromatography (TLC), high-performance thin layer chromatography (HPTLC), high-performance liquid chromatography (HPLC), gas chromatography (GC) and hyphenated techniques like HPLC-MS, GC–MS NMR etc. (Mukherjee et al., 2008). All these techniques can help to quantify the phytochemicals present in complex mixtures of Ayurvedic products. Method development and analytical evaluation of chemical constituents of Ayurvedic formulation through chemo-profiling is a challenging task. Several aspects of quality control, standardization and validation of Ayurvedic medicine has been presented in Fig. 4. The main aim of the quality evaluation is to ensure a predefined amount of quantity, quality and therapeutic effect of Ayurvedic products (Mukherjee et al., 2009).

3.3. Chemo-profiling and standardization of Ayurvedic medicine

Chemo-profiling should be extended to several groups or classes of constituents and a chemical fingerprint can be established to evaluate the phytoconstituents present therein. Chemical fingerprints can be used to authenticate plant material, identify the quantification of active compounds and relate the chemical composition to biological activity for product standardization (Mukherjee and Houghton, 2009). A cautionary principle also applies to standardization; i.e if manufacturers do not use the same markers or methods to obtain standardization. Thus, similar products may not be standardized to the same marker compound or if to the same marker, to different levels. Standardization and validation requires control of both raw material quality and manufacturing processes to final product (Mukherjee et al., 2011). Marker analysis of traditional medicines is a critical and essential matter to be considered in assuring the therapeutic efficacy, drug metabolism and pharmacokinetics (DMPK), safety, shelf life and stability study to rationalize their use in the healthcare. This is an integral part of traditional medicine, which ensures that it delivers the required quantity of quality medicament. Marker profiling is the thrust area for traditional formulations like churnas, bhasmas, liquid orals, lehas, etc. (Mukherjee, 2001). The finger printing and marker compound analysis are nowadays gaining importance for the standardization of Ayurvedic formulations (Fig. 4). However, for most Ayurvedic medicines, the therapeutic components have not been fully elucidated or easily monitored. Chemo-profiling of some promising leads from Ayurvedic medicines are given in the Table 2.

Many compounds have been derived from the Ayurveda which have been mentioned in the ancient texts of the Indian system of medicine for the treatment of several ailments. Combining the strengths of the knowledge base of traditional systems of Ayurveda with the dramatic power of combinatorial approaches and high throughput screening will be helpful in development of new lead molecules (Patwardhan, et al., 2004). A considerable amount of research on pharmacognosy, chemistry, pharmacology and clinical therapeutics has been carried out on Ayurvedic medicinal plants. Numerous molecules have come out of Ayurvedic experimental base, including Rauwolfia alkaloids (reserpine) for hypertension, psoralens for vitiligo, Holarrhena alkaloids in amoebiasis, guggulipid (guggulsterons) from *Commiphora mukul* Hook as hypolipidemic agents, L-dopa from *Mucuna pruriens* L. (D.C.) for Parkinson's disease, piperine as bioavailability enhancers from *Piper nigum* L., baccosides from *Bacopa monnieri* L. for mental retention, picrosides for hepatic protection, phyllanthins as antivirals, curcumines for inflammation, withanolides from *Withania somnifera* L. and many other steroidal lactones and their glycosides as immunomodulators from *Asparagus racemosus* Willd., *Tinospora cordifolia* Willd. etc. (Patwardhan, 2005). There are growing incidences where the old molecules are finding new application though better understanding of traditional knowledge and clinical observations. For instance, forskolin an alkaloid isolated from *Coleus forskohlii* Willd. by Central Drug Research Institute (CDRI), CSIR, Lucknow is used for obesity and atherosclerosis. Potent antimicrobial, antirheumatic and cyclooxygenase inhibitory activities of phenolics, catechols and flavonoids from an important Ayurvedic plant *Semecarpus anacardium* L. F. have been reported as promising leads (Patwardhan and Vaidya, 2010).

Please cite this article as: Mukherjee, P.K., et al., Development of Ayurveda – Tradition to trend. Journal of Ethnopharmacology (2016), http://dx.doi.org/10.1016/j.jep.2016.09.024

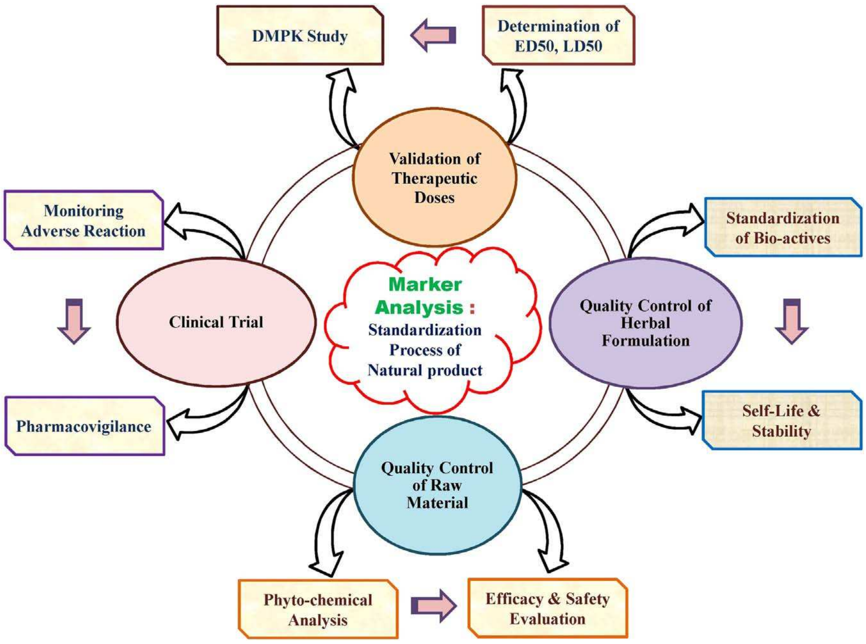


Fig. 4. Approaches for quality control, standardization and validation of Ayurvedic medicine.

4. Evidence based Ayurvedic drug development

कालबुद्धीन्द्रयािर्थानां योगो मथि्या न चातिचा।.

द्वयाश्रयाणांव्याधीनांत्रविधिोहेतुसङ्ग्रहः॥.

Ch. Su. – 1/54.

[*Perverted, negative and excessive use of time, intelligence and. sense objects is the threefold cause of both psychic and somatic disorders*].

Ayurveda focuses on disease prevention and optimizing vitality as much as on removing an illness. It is a holistic approach to health that includes balance of the three biological factors known as *tridosha* (*Vata, Pitta* and *Kapha*) together with aspects of life in a philosophy where *satma* (mind), *atma* (spirit), *sharira* (physical) and *indriya* (sense organs or intellect) are considered to be an integrated whole. Ayurveda has expanded these basic concepts to correlate with human physiological factors like *dosha, dhatu* (body tissues), *mala* (excretory product), *agni* (metabolic factor), *ama* (unassimilated toxins) and *srotasa* (major body channels). These factors govern the body functions and pathophysiology. The concepts are being used and developed since ancient times which led to the development of different disciplines as clinical modalities and knowledge-based strategies of drug development which has been detailed earlier (see Fig. 1). Traditional knowledge and concepts when integrated with modern scientific approach present a unique model of drug development and healthcare. This evidence-based integrated model of Ayurveda present opportunities for developing safe and effective treatment choices for the future as explained in Fig. 5.

Charaka Samhita and *Sushruta Samhita* are the main classics of Ayurveda, which contain detailed descriptions of >700 medicinal herbs and >8000 formulations. Evidence-based studies should be focused on traditional systems of medicines dealing with both crude drugs and purified phyto-molecules (Mukherjee, 2010). Researchers have also proposed a way forward to address the changing scenario in the development and promotion of Ayurveda (Mukherjee et al., 2012). Multi-target approaches are in the mainstream of renewed interest in multi-ingredient synergistic formulations (Zimmermann et al., 2007). The multi-target approaches deals with the single components of individual or mixtures of herbs which affect not only single site (one target) but also several sites (multi-targets). The therapeutic effects may be observed in an agonistic and synergistic manner. There are several drug targets for the therapeutic activities such as enzymes, proteins, receptors, substrates, metabolites and ion channels, transport proteins, DNA/RNA, ribosomes, monoclonal antibodies etc. (Wagner and Ulrich-Merzenich, 2009). The most of the Ayurvedic drug and phytomedicines are available in the market as 'polyherbal formulation' which is the whole extracts of herbs. It is generally believed that these formulations may exert their therapeutic activity in a synergistic way. Although in the present scenario, the isolation of single constituents from herbs is a prime focus but in case of poly-formulation, the therapeutic efficacy is a result of combined effect of the constituents at low concentrations present in the herbs (Williamson, 2001; Mukherjee et al., 2011).

Ayurvedic texts describes about thousands of single or polyherbal formulations. These are rationally designed and are in therapeutic use for a long time. Pharmaco-epidemiological evidence must be generated to support their safety and efficacy. Systematic data mining of the huge existing formulations database can certainly expedite drug discovery processes to identify potent molecules. The US Food and Drug Administration (FDA) and few other regulators do have very practical guidelines for 'botanical drug development and herbals' that are no more restricted to nutraceuticals. The Ayurvedic Formulary gives thousands of such multi-ingredient preparations and an excellent rationale for such formulations in the Ayurvedic classics. For example, one of the pioneering series of clinical studies on Ayurvedic antiarthritis formulation known as a Rheumayog which confirmed the disease modifying activity in comparison with modern drug, Auranofin (Chandrasekaran et al., 1994). The successful attempt to design multi-ingredient formulation, Artrex to treat rheumatoid arthritis and osteoarthritis (OA) has been validated by a well-designed randomized, double-blind, placebo-controlled clinical trial. The formulation shows its effects when compared to a combination of NSAIDs and disease-modifying anti-rheumatic drugs. The product has been patented in India and the United States (Patwardhan, 1996).

The Council for Scientific and Industrial Research (CSIR) under the New Millennium Indian Technology Leadership Initiative

Table 2
Promising leads inspired from Ayurvedic herbs.

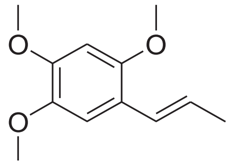
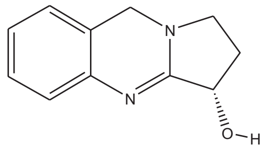
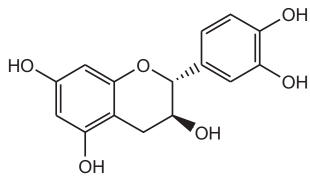
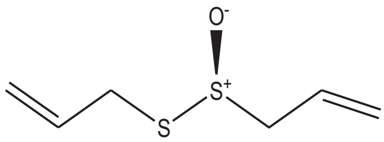
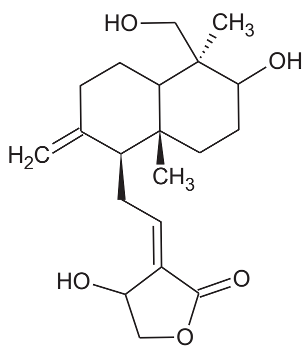
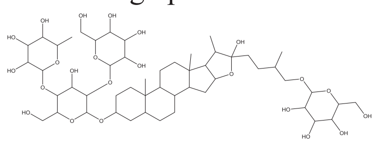
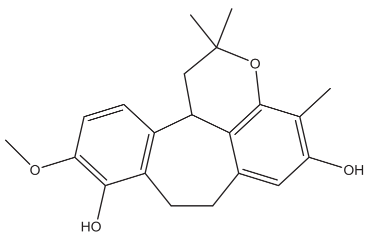
Ayurvedic plant	Common name	Parts used	Major phytoconstituents	Therapeutic activity	Reference
Acorus calamus L. (Acoraceae)	Vacha	Rhizome	 α-Asarone	Memory disorder, learning performance, stomachache, anti-aging, anticholinergic activity	Mukherjee et al., 2007
Adhatoda vasica Nees (Acanthaceae)	Vasaka	Fresh, dried mature leaves	 Vasicine	Antibacterial, bronchodilator, antiasthmatic, antiulcer, anticancer	Singh et al., 2011
Albizia lebbeck L. (Leguminosae)	Shirisha	Stem bark	 Catechin	Anti-allergic, antihistaminic	Venkatesh et al., 2010
Allium sativum L. (Liliaceae)	Lahsuna	Bulb	 Allicin	Hypolipidaemic, antiatherosclerotic, hypoglycaemic, anticoagulant, anti-hypertensive, antimicrobial, anticancer, antidote (for heavy metal poisoning), hepatoprotective and immunomodulatory	Banerjee et al., 2003
Andrographis paniculata Nees (Acanthaceae)	Kalmegh	Leave	 Andrographolide	Treatment of fever, inflammation, common cold, upper respiratory tract infection, omitingi, pharyngitis, laryngitis, pneumonia, tuberculosis, pyelonephritis and hepatic disorder	Maiti et al., 2010
Asparagus racemosus Willd (Liliaceae)	Vasaka	Root	 Shatavarins I  Racemosol	Antibacterial, bronchodilatory, antiasthmatic, antiulcer, anticancer	Alok et al., 2013

Table 2 (*continued*)

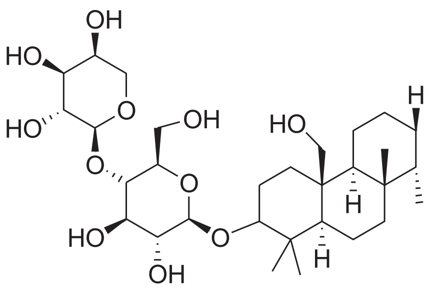
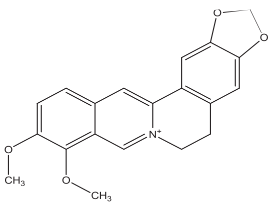
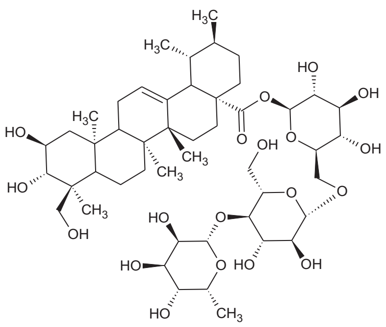
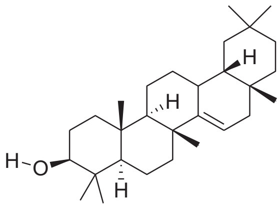
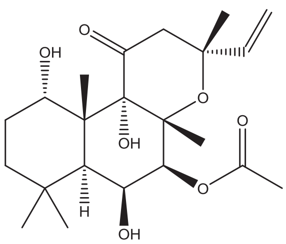
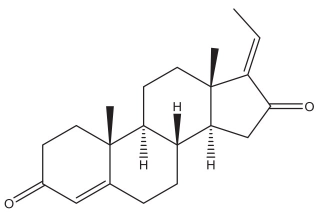
Ayurvedic plant	Common name	Parts used	Major phytoconstituents	Therapeutic activity	Reference
Bacopa monnieri L. (Scrophulariaceae)	Mandukaparni	Whole plant	 Bacoside	Used as memory enhancer, brain tonic, antiasthmatic and antipyretic	Ganzera et al., 2004
Berberis aristata D.C. (Berberidaceae)	Daruharidra	Dried stem	 Berberine	Hepatoprotective, antidiarrhoeal, cardiotonic, antidiabetic,	Potdar et al., 2012
Centella asciatica L. (Apiaceae)	Brahmi	Leave	 Asiaticoside	Anti-wrinkle, used in wound healing and antihistimincs	Nema et al., 2013
Clitoria ternatea L. (Fabaceae)	Aparajita	Root, flower, leave	 Taraxerol	Antimicrobial, antipyretic, anti-inflammatory, analgesic, diuretic, local anaesthetic, antidiabetic, insecticidal, anti-platelet	Mukherjee et al., 2008
Coleus forskohlii Willd. (Labiatae)	Pashan Bhedi	Root	 Forskolin	Used in hypertension, congestive heart failure, intestinal disturbance, respiratory disorders, insomnia, convulsions, liver fatigue, glaucoma, Alzheimer's disease andfor prevention of cancer metastases	Harde et al., 2014
Commiphora wightii Arn. and *Commiphora mukul* Hook. (Burseraceae)	Guggul	Exudates	 Guggulsterone E	Antihyperlipidemic, cardioprotective, antiwrinkle, antimicrobial, anti-inflammatory, analgesic, hepatoprotective	Bhatia et al., 2015;Shen et al., 2012

Table 2 (*continued*)

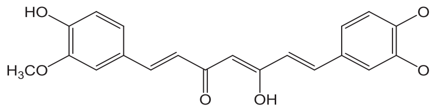
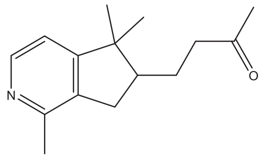
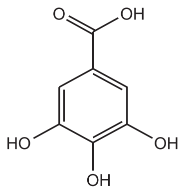
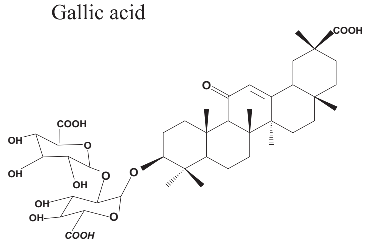
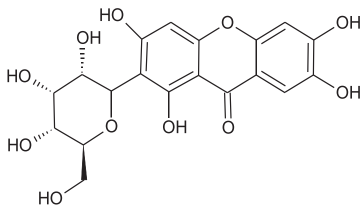
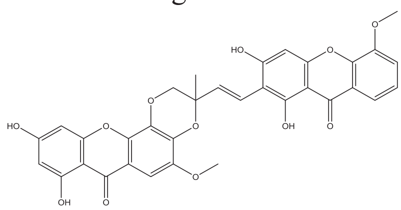
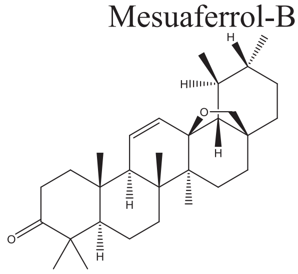
Ayurvedic plant	Common name	Parts used	Major phytoconstituents	Therapeutic activity	Reference
Curcuma longa L. (Zingiberaceae)	Haldi	Rhizome	 Curcumin	Antitumour, antioxidant, antiarthritic, antiamyloid, anti-ischemic, antiinflammatory	Gantait et al., 2011
Cyperus rotundus L. (Cyperaceae)	Nagarmotha	Dried rhizome	 Rotundine A	Analgesic, antiandrogenic, antimicrobial, anticonvulsant, antidiabetic, antilipidemic, neuroprotective, nootropic, lactogenic, hepatoprotective	Pirzada et al., 2015
Emblica officinalis Gaertn. (Euphorbiaceae)	Amlaki	Fruit	 Gallic acid	Hepatoprotective, anti-oxidant, antidiabetic, anti-tumor, immunomodulatory	Ponnusankar et al., 2011
Glycyrrhiza glabra L. (Leguminosae)	Mulethi	Root and rhizome	 Glycyrrhizin	Anti-inflammatory and antiulcer, hepatoprotective, antiallergic, anti-arthritic, anti-arrhythmic, antibacterial, antiviral and antiasthmatic	Harwansh et al., 2011
Mangifera indica L. (Anacardiaceae)	Aam	Leaf, bark, fruits	 Mangiferin	Antioxidant, hypolipidemic, antidiabetic, antiatherogenic, liver problem	Rai et al., 2007
Mesua ferrea L. (Guttiferae)	Nagkesara	Dried stamen	 Mesuaferrol-B	Antiasthmatic, anti-inflammatory, estrogenic, antiplasmodial, antimicrobial	Chanda et al., 2013
Momordica charantia L. (Cucurbitaceae)	Karela	Fresh fruit	 Momordicin	Antidiabetic, antiviral, antibacterial, anticancer	Grover and Yadav., 2004

Table 2 (*continued*)

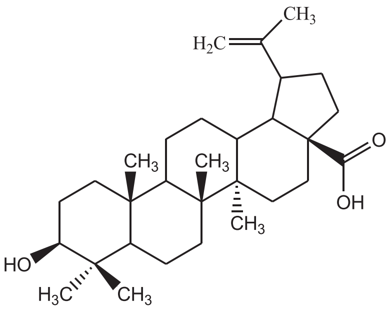
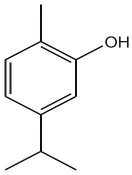
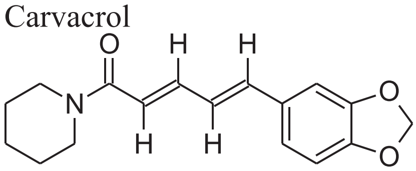
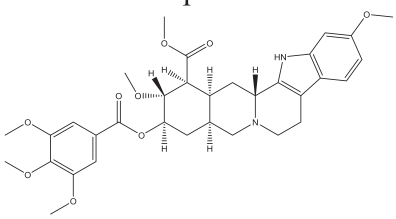
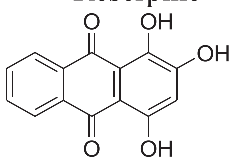
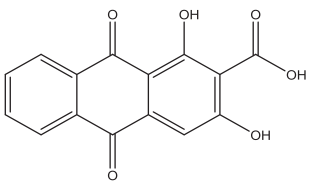
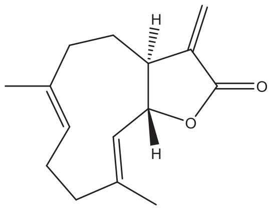
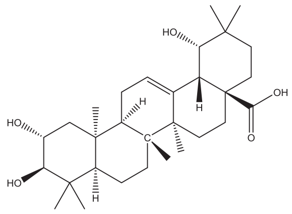
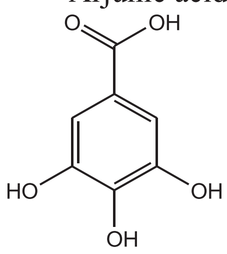
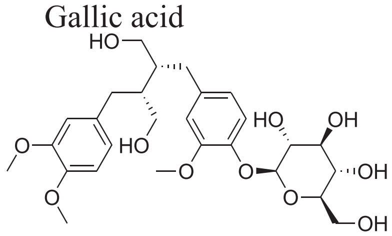
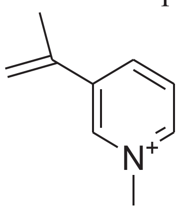
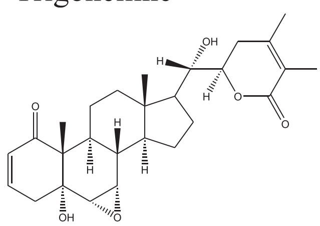
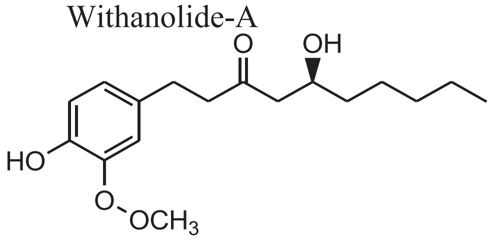
Ayurvedic plant	Common name	Parts used	Major phytoconstituents	Therapeutic activity	Reference
Nelumbo nucifera Gaertn. (Nymphaceae)	Kamal	Rhizome	 Betulinic acid	Used in pharyngopathy, pectoralgia, leucoderma, strangury, dysentery, cough, haematemesis, tissue inflammation, cancer, skin diseases and diabetes	Mukherjee et al., 2010
Ocimum sanctum L. (Lamiaceae)	Tulasi	Leave	 Carvacrol	Antioxidant, antimicrobial, antidiabetic, anticancer, antifertility, hepatoprotective, antihypertensive, analgesic, adaptogenic	Chaudhary et al., 2014
Piper longum L. and *Piper nigrum* L. (Piperaceae)	Pipar and kalimirch	Fruit	 Piperine	Antiageing, revitalizing, memory enhancing, adaptogenic, anti-diarrhoeal, antispasmodic, immunomodulatory, remedies for cough, cold, fever, asthma and other respiratory problems	Harwansh et al., 2014
Rauwolfia serpentina L. (Apocyanaceae)	Sarpagandha	Root	 Reserpine	Antidepressant, anxiolytic	Vakil, 1955
Rubia cordifolia L. (Rubiaceae)	Manjistha	Root	 Purpurin  Manjistin	Used in skin ailments, wounds, ulcers, acne, antioxidant, radioprotective, hepato-protective, tyrosinase inhibitor	Biswas et al., 2015
Saussurea lappa C.B. (Compositae)	Kustha	Dried roots	 Costunolide	Anticancer, hepatoprective, antiulcer, immunomodulator, anticonvulsant	Zahara et al., 2014

Table 2 (*continued*)

Ayurvedic plant	Common name	Parts used	Major phytoconstituents	Therapeutic activity	Reference
Terminalia arjuna Wight & Arn. (Combretaceae)	Arjuna	Stem bark	 Arjunic acid	Cardiprotective, antihyperlipidemic, antiatherogenic	Dwivedi and Chopra, 2014
Terminalia belerica Roxb. and *Terminalia chebula* Retz.	Bibhitaki and haritaki	Fruit	 Gallic acid	Anti-atherosclerotic, hepatoprotective, cardioprotective, cytoprotective, cardiotonic, antimutagenic and antifungal	Ponnusankar et al., 2011
Tinospora cordifolia Willd. (Menispermaceae)	Guduchi	Stem bark	 Tinosporoside A	Anticancer, antidiabetic, anti-inflammatory, hypolipidemic	Shirolkar et al., 2013
Trigonella foenum-graecum L. (Fabaceae)	Methi	Fruit (Seed)	 Trigonelline	Anti-cholinesterase, antidiabetic, antiulcer, wound healing, CNS stimulant, anti-inflammatory and antipyretic immunomodulatory, antioxidant	Kumar et al., 2010
Withania somnifera L. (Solanaceae)	Ashwagandha	Leaves, roots	 Withanolide-A	Adaptogenic, anticancer, anti-convulsant, immunomodulatory, antioxidative and neurological	Chatterjee et al., 2010
Zingiber officinale Rosc. (Zingiberaceae)	Sunthi	Rhizome	 Gingerol	Antiviral (hRSV), omitingicalry, bronchitis and other respiratory tract infections	Harwansh et al., 2014

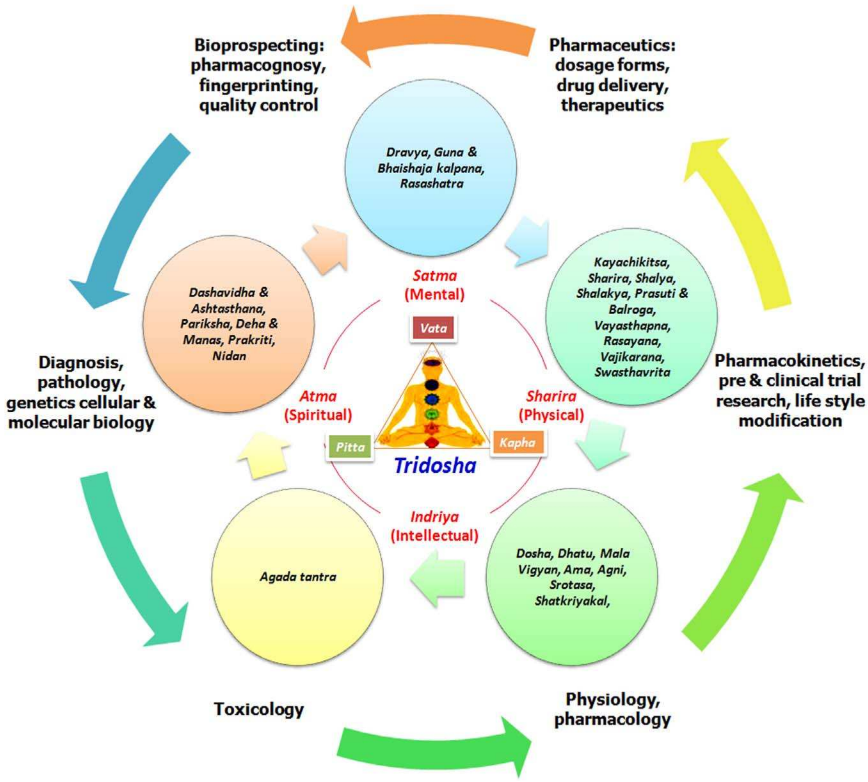


Fig. 5. Evidence based approaches of drug development from Ayurveda.

(NMITLI) program of the Government of India supported an Ayurveda-based herbal drug development project. This project was a team work comprising six research institutes, five clinical centres and six industry collaborators. The clinical team developed integrative protocols and appropriate research methodologies for evidence-based Ayurveda and botanical drug development (Chopra et al., 2010). Two formulations, which performed better than placebo and standard drug glucosamine in preliminary trials, were then taken up for further mechanistic studies (Chopra et al., 2011). All the formulations prepared for clinical trials were manufactured by following USFDA guidance to Industry for botanical drugs.

Rasayana therapy is a comprehensive regimen that encompasses the concept of "rejuvenation" of mental and physical health. In Ayurveda "*rasayana*" refers to acquisition, movement, or circulation of nutrition needed to the body tissue (known as dhatus in Ayurveda). Classical formulations such as *chyawanprasha* and *brahmi rasayana* are quite useful for all age groups. Rasayanas are a group of medicinal plants described in Ayurveda for rejuvenation, anti-aging and immunomodulatory properties. Several studies on *rasayana* have made outstanding contributions to ethnopharmacology (Jadhav and Patwardhan, 2003). Researchers have studied some "*rasayana*" plants, including *Withania somnifera* L. (ashwagandha), *Asparagus racemosus* Willd. (Shatavari), *Tinospora cordifoloia* Willd. (Guduchi), *Emblica officinalis* Gaertn. (Amalaki), and *Semecarpus anacardium* L. F. (Bhallataka), for several therapeutic activity for various standardized extracts and formulations. In one particular study, it has been indicated that Ashwagandha is a better and safer drug than Ginseng (Grandhi et al., 1994). Triphala is also an Ayurvedic *rasayana* of three popular fruits viz. *E. officinalis, Terminalia chebula* Retz. and *Terminalia belerica* Roxb. in equal proportions as described in Ayurvedic Formulary of India (AFI) (Anonymous, 2002). Triphala is mentioned for its several therapeutic benefits in Ayurvedic literature mainly bowel regulator and tonic. It is also used as cleanser, blood purifier, eyewash to counter many eye ailments e.g. conjunctivitis, redness and soreness of eyes (Mukherjee et al., 2006). This *rasayana* can balance the '*Tridoshas*' (three biological humors viz. *Vata* (space and air), *Pitta* (fire and water), and *Kapha* (water and earth) (Debnath et al., 2015). Trikatu is a very well known 'rasayana' in Ayurveda which consists of three Ayurvedic plants namely *Piper longum* L., *Piper nigrum* L. and *Zingiber officinale* Rosc. in equal ratio. Trikatu has been prescribed for several ailments including cough, cold, fever, asthma, respiratory problems, and improvement of digestive disorders. Safety profiles of the Triphala and Trikatu has been established through CYP450 enzyme inhibition assay models (Ponnusankar et al., 2011; Harwansh et al., 2014).

5. Challenges in development of Ayurvedic formulation

Development of suitable dosage forms of Ayurvedic or herbal drugs is still a challenging task. Some primary constraints for developing an appropriate delivery system are limited solubility and permeability of herbal drugs through biological membranes. Therefore, it produces little therapeutic efficacy and bioavailability. Research is now being concurrently conducted on basic as well as applied fields of herbal medicines, and this has created the need for studies in the delivery system of herbal drugs for maximum bioavailability (Mukherjee et al., 2009). The biological half-life ($t_{1/2}$) has an immense role in the therapeutic efficacy and potency of drug molecules at the site where it is administered. If the drugs have shorter $t_{1/2}$ than it possesses low bioavailability as compared to higher $t_{1/2}$ values. The biological membrane permeability offers more activity for a lipophilic drug. Hence, its chances of availability in the blood/plasma are more in comparison to the hydrophilic drug. However, number of the active pharmaceutical ingredients (API) obtained from high-throughput screening are poorly soluble. Lower bioavailability results from poor solubility and incomplete dissolution in vivo. These are often holding back continuous development and coming into the market of some promising new chemical entities (NCEs). Hence, innovative formulation approaches are gaining attentions for acquiring better therapeutic efficacy through improved bioavailability (Rahman et al., 2013). In this context, novel drug delivery systems (NDDS) efficiently can overcome some drawback associated with the Ayurvedic drug. Thus, the pharmaceutical formulation for delivery of phytoconstituents is of premier importance. The advancement of NDDS is providing a large platform for the Ayurvedic drug to improve their therapeutic activity (Mukherjee et al., 2015).

6. Conclusion

Ayurvedic drug is mentioned in the ancient text of traditional Indian system of medicine that could be more explored and validated in association with the modern scientific technologies. Therapeutic efficacy of Ayurvedic medicine may be enhanced to high quality which can be achieved by evaluation of identity, purity, safety, stability, physical and biological properties. Chemoprofiling, standardization and biological study are very essential for the quality evaluation as well as to maintain scientific documentation for validation of Ayurvedic medicine. Ayurveda is a self sustaining system of traditional medicine. The importance of the Ayurvedic knowledge in drug discovery and healthcare is expanding ever since. As more and more scientific validation of this ancient medical system is emerging, new avenues are opening up for scientific exploration. The integrated evidence based research on Ayurveda thus provides opportunities for a better healthcare system serving millions with a hope for an efficient and safe therapeutics.

Conflict of interest

Authors declare no conflict of interest.

Acknowledgement

The authors are thankful to the Department of Biotechnology, Government of India, New Delhi, for financial support through Tata Innovation Fellowship (D.O. No. BT/HRD/35/01/04/2014).

We would like to dedicate this article in memory of beloved Prof. P. K. Debnath, who as an Ayurvedic expert guided us in different aspects of our work; unfortunately, he passed away on April 25, 2015.

References

Alok, S., Jain, S.K., Verma, A., Kumar, M., Mahor, A., Sabharwal, M., 2013. Plant profile, Phytochemistry and Pharmacology of *Asparagus racemosus* (Shatavari): a review. Asian Pac. J. Trop. Dis. 3, 242–251.

Anonymous, 2002. The Ayurvedic Formulary of India, Part II. Department of Indian System of Medicine and Homeopathy, New Delhi.

Anonymous, 2012. Ayurveda the science of life, Department of AYUSH, Ministry of Health and Family Welfare, Govt. of India, New Delhi.

Bastos, C., 2009. Book review: refiguring unani tibb: plural healing in late colonial India. Med. Hist. 53 (4), 594–595.

Banerjee, S.K., Mukherjee, P.K., Maulik, S.K., 2003. Garlic as an antioxidant: the good, the bad and the ugly. Phytother. Res. 17, 97–106.

Bhatia, A., Bharti, S.K., Tripathi, T., Mishra, A., Sidhu, O.P., Roy, R., Nautiyal, C.S., 2015. Metabolic profiling of *Commiphora wightii* (guggul) reveals a potential source for pharmaceuticals and nutraceuticals. Phytochemistry 110, 29–36.

Biswas, R., Mukherjee, P.K., Dalai, M.K., Mandal, P.K., Nag, M., 2015. Tyrosinase inhibitory potential of purpurin in *Rubia cordifolia* - a bioactivity guided approach. Ind. Crops Prod. 74, 319–326.

Chanda, S., Rakholiya, K., Parekh, J., 2013. Indian medicinal herb: antimicrobial efficacy of *Mesua ferrea* L. seed extracted in different solvents against infection causing pathogenic strains. J. Acute Dis., 277–281.

Chandra, T.R., Suresh, C., Nigamanand, B., Deep, K.G., 2014. Guidelines for collection of raw drugs and characteristics of collected material w.s.r. to Ayurveda. Unique J. Ayurvedic Herb. Med. 2, 4–7.

Chandrasekaran, A.N., Porkodi, R., Radhamadhavan, Parthiban, M., Bhatt, N.S., 1994. Studies on Ayurvedic drugs in rheumatoid arthritis comparison with Auranofin. Indian Pract. XLVII (6), 489–502.

Chatterjee, S., Srivastava, S., Khalid, A., Singh, N., Sangwan, R.S., Sidhu, O.P., Roy, R., Khetrapal, C.L., Tuli, R., 2010. Comprehensive metabolic fingerprinting of *Withania somnifera* leaf and root extracts. Phytochemistry 71, 1085–1094.

Chaudhary, S.K., Mukherjee, P.K., Maity, N., Bhadra, S., Saha, B.P., 2014. *Ocimum sanctum* L. a potential angiotensin converting enzyme (ACE) inhibitor useful in hypertension. Indian J. Nat. Prod. Resour. 5, 83–87.

Chopra, A., Saluja, M., Tillu, G., 2010. Ayurveda-modern medicine interface: a critical appraisal of studies of Ayurvedic medicines to treat osteoarthritis and rheumatoid arthritis. J. Ayurveda Integr. Med. 17, 190–198.

Chopra, A., Saluja, M., Tillu, G., Venugopalan, A., Sarmukaddam, S., Raut, A.K., Bichile, L., Narsimulu, G., Handa, R., Patwardhan, B., 2011. A randomized controlled exploratory evaluation of standardized Ayurvedic formulations in symptomatic osteoarthritis knees: A government of India NMITLI project. Evid. Based Complement Altern. Med. 724291, 1–12.

Cordell, G.A., 2011. Phytochemistry and traditional medicine – a revolution in process. Phytochem. Lett. 4, 391–398.

Debnath, P.K., Banerjee, S., Debnath, P., Mitra, A., Mukherjee, P.K., 2015. Ayurveda – opportunities for developing safe and effective treatment choices for the future. In: Mukherjee, P.K (Ed.), Evidence-Based Validation of Herbal Medicines. Elsevier, Amsterdam, pp. 427–454.

Dwivedi, S., Chopra, D., 2014. Revisiting *Terminalia arjuna* – an ancient cardiovascular drug. J. Tradit. Complement. Med. 4, 224–231.

Folashade, K.O., Omoregie, E.H., Ochogu, A.P., 2012. Standardization of herbal medicines – a review. Int. J. Biodivers. Conserv. 4, 101–112.

Gangadharan, N., 1982. The state of ayurveda in the eighteenth & nineteenth centuries. Indian J. Hist. Sci. 17, 154–163.

Gantait, A., Barman, T., Mukherjee, P.K., 2011. Validated method for estimation of curcumin in turmeric powder. Indian J. Tradit. Knowl. 10, 247–250.

Ganzera, M., Gampenrieder, J., Pawar, R.S., Khan, I.A., Stuppner, H., 2004. Separation of the major triterpenoid saponins in *Bacopa monnieri* by high-performance liquid chromatography. Anal. Chim. Acta 516, 149–154.

Grandhi, A., Mujumdar, A.M., Patwardhan, B., 1994. A comparative pharmacological investigation of ashwagandha and ginseng. J. Ethnopharmacol. 44, 131–135.

Grover, J.K., Yadav, S.P., 2004. Pharmacological actions and potential uses of *Momordica charantia*: a review. J. Ethnopharmacol. 93, 123–132.

Harde, S.M., Lonkar, S.L., Degani, M.S., Singhal, R.S., 2014. Ionic liquid based ultrasonic-assisted extraction of forskolin from *Coleus forskohlii* roots. Ind. Crops Prod. 61, 258–264.

Harwansh, R.K., Mukherjee, K., Bhadra, S., Kar, A., Bahadur, S., Mitra, A., Mukherjee, P.K., 2014. Cytochrome P450 inhibitory potential and RP-HPLC standardization of trikatu – a rasayana from Indian Ayurveda. J. Ethnopharmacol. 153, 674–681.

Harwansh, R.K., Patra, K.C., Pareta, S.K., Singh, J., Biswas, R., 2011. Pharmacological studies of *Glycyrrhiza glabra*: a review. Pharmacologyonline 2, 1032–1038.

Jadhav, R.B., Patwardhan, B., 2003. Anti-anxiety activity of *Celastrus paniculatus*. Indian J. Nat. Prod. 19, 16–19.

Jaggi, O.P., 1973. Medicine in Medieval India, pp. 209–222.

Krishnamachary, B., Rajendran, N., Pemiah, B., Krishnaswamy, S., Krishnan, U.M., Sethuraman, S., Sekar, R.K., 2012. Scientific validation of the different purification steps involved in the preparation of an Indian Ayurvedic medicine, *Lauha bhasma*. J. Ethnopharmacol. 142, 98–104.

Kumar, N.S., Mukherjee, P.K., Bhadra, S., Saha, B.P., 2010. Acetylcholinesterase enzyme inhibitory potential of standardized extract of *Trigonella foenum-graecum* L and its constituents. Phytomedicine 17, 292–295.

Kutumbiah, P., 1962. Reissued 1999. Ancient Indian Medicine. Orient Longman Ltd, Chennai.

Lavekar, G.S., 2006. Scientific Validation of Drug Development and Clinical Research in Ayurveda. AYU. PM70, pp. 66–85.

Maiti, K., Mukherjee, K., Murugan, V., Saha, B.P., Mukherjee, P.K., 2010. Enhancing bioavailability and hepatoprotective activity of andrographolide from *Andrographis paniculata*, a well known medicinal food, through its herbosome. J. Sci. Food Agr. 90, 43–51.

Misra, R.P., 2007. Geography of Health: A Treatise on Geography of Life and Death in India. New Concept Publishing Company, Delhi.

Mukherjee, P.K., Houghton, P.J., 2009. The worldwide phenomenon of increased use of herbal products: opportunity and threats. In: Mukherjee, P.K., Houghton, P.J. (Eds.), Evaluation of Herbal Medicinal Products - Perspectives on Quality, Safety and Efficacy. Pharmaceutical Press, London, pp. 3–12.

Mukherjee, D., Kumar, N.S., Khatua, T., Mukherjee, P.K., 2010. Rapid validated HPTLC method for estimation of betulinic acid in *Nelumbo nucifera* (Nymphaeaceae) rhizome extract. Phytochem. Anal. 21, 556–560.

Mukherjee, P.K., Venkatesh, M., Gantait, A., 2010. Comprehensive Natural Products II. In: Mander, L., Lui, H.W. (Eds.), Chemistry and Biology. Elsevier, Oxford, pp. 479–507.

Mukherjee, P.K., Bahadur, S., Harwansh, R.K., Chaudhary, S.K., 2014. Shifting paradigm for validation of medicinal plants in Indian traditional medicine. Indian Drugs 51, 5–14.

Mukherjee, P.K., Bahadur, S., Harwansh, R.K., Nema, N.K., Bhadra, S., 2013. Development of traditional medicines: globalizing local knowledge or localizing global technologies. Pharma Times 45, 39–42.

Mukherjee, P.K., 2001. Evaluation of Indian traditional medicine. Drug Inf. J. 35, 623–631.
Mukherjee, P.K., 2005. Promotion and development of botanicals with international coordination. In: Mukherjee, P.K. (Ed.), Promotion and Development of Botanicals with International Coordination: Exploring Quality, Safety, Efficacy and Regulations. Allied Book Agency, Kolkata, pp. 1–9.
Mukherjee, P.K., Bahadur, S., Chaudhary, S.K., Kar, A., Mukherjee, K., 2015. Quality related safety issue-evidence-based validation of herbal medicine farm to pharma. In: Mukherjee, P.K. (Ed.), Evidence Based Validation of Herbal Medicine. Elsevier, Amsterdam, pp. 1–28.
Mukherjee, P.K., Harwansh, R.K., Bahadur, S., Banerjee, S., Kar, A., 2016. Evidence based validation of Indian traditional medicine - Way forward. World J. Tradit. Chin. Med. 2, 48–61.
Mukherjee, P.K., Kumar, V., Kumar, N.S., Heinrich, M., 2008. The Ayurvedic medicine *Clitoria ternatea* from traditional use to scientific assessment. J. Ethnopharmacol. 120, 291–301.
Mukherjee, P.K., Kumar, V., Mal, M., Houghton, P.J., 2007. *Acorus calamus*: scientific validation of Ayurvedic tradition from natural resources. Pharm. Biol. 45, 651–666.
Mukherjee, P.K., Nema, N.K., Venkatesh, P., Debnath, P.K., 2012. Changing scenario for promotion and development of Ayurveda – way forward. J. Ethnopharmacol. 143, 424–434.
Mukherjee, P.K., Ponnusankar, S., Pandit, S., Hazam, P.K., Ahmmed, M., Mukherjee, K., 2011. Botanicals as medicinal food and their effects on drug metabolizing enzymes. Food Chem. Toxicol. 49, 3142–3153.
Mukherjee, P.K., Saha, B.P., 2003. Quest for GMP for the production of quality botanicals. In: Mukherjee, P.K., Verpoorte, R. (Eds.), GMP for Botanicals regulatory and Quality Issues on phytomedicine. Eastern Publishers, New Delhi, pp. 165–190.
Mukherjee, P.K., Venkatesh, M., Maiti, K., Mukherjee, K., Saha, B.P., 2009. Value added herbal drug delivery systems - perspectives and developments. Indian J. Pharm. Educ. Res. 43, 329–337.
Mukherjee, P.K., Wahile, A., 2006. Integrated approaches towards drug development from Ayurveda and other Indian system of medicines. J. Ethnopharmacol. 103, 25–35.
Nema, N.K., Maity, N., Sarkar, B.K., Mukherjee, P.K., 2013. Matrix metalloproteinase, hyaluronidase and elastase inhibitory potential of standardized extract of *Centella asiatica*. Pharm. Biol. 51, 1182–1187.
Panikkar, K.N., 2002. Culture, Ideology, Hegemony: Intellectuals and Social Consciousness in Colonial India. Anthem Press, London.
Patwardhan, B., 2005. Ethnopharmacology and drug discovery. J. Ethnopharmacol. 100, 50–52.
Patwardhan, B., 1996. Method of Treating Musculoskeletal Disorders and a Novel Composition Therefore. United States Patent No. 5494668, Licensed to BioVed Inc, San Jose, CA, USA.
Patwardhan, B., Vaidya, A.D.B., 2010. Natural Product drug discovery: accelerating the clinical candidate development using reverse pharmacology approaches. Indian J. Exp. Biol. 48, 220–227.
Patwardhan, B., Vaidya, A.D.B., Chorghade, M., 2004. Ayurveda and natural product drug discovery. Curr. Sci. 86, 789–799.
Pirzada, A.M., Ali, H.H., Naeem, M., Latif, M., Bukhari, A.H., Tanveer, A., 2015. *Cyperus rotundus* L.: traditional uses, phytochemistry, and pharmacological activities. J. Ethnopharmacol. 174, 540–560.
Ponnusankar, S., Pandit, S., Babu, R., Bandyopadhyay, A., Mukherjee, P.K., 2011. Cytochrome P450 inhibitory potential of Triphala – a rasayana from Ayurveda. J. Ethnopharmacol. 133, 120–125.
Potdar, D., Hirwani, R.R., Dhulap, S., 2012. Phyto-chemical and pharmacological applications of *Berberis aristata*. Fitoterapia 83, 817–830.
Rahman, M.A., Hussain, A., Iqbal, Z., Harwansh, R.K., Singh, L.R., Ahmad, S., 2013. Nanosuspension: a potential nanoformulation for improved delivery of poorly bioavailable drug. Micro Nanosyst. 5, 273–287.
Rai, S., Basak, S., Mukherjee, K., Saha, B.P., Mukherjee, P.K., 2007. Oriental medicine *Mangifera indica*. Orient Pharm. Exp. Med. 7, 1–10.
Ray, P.C., 1956. History of Chemistry in Ancient and Medieval India Incorporating the History Of Hindu Chemistry. Indian Chemical Society, Calcutta.
Roy, A., Debnath, P.K., 2000. Rogavinischaya (Bengali Version). J.B. Roy State Ayurvedic Medical College & Hospital, Kolkata.
Sarkar, T., 2001. Hindu Wife, Hindu Nation: Community, Religion and Cultural Nationalism. Indiana University Press, Bloomington, Indiana.
Sharma, P.V. (Ed.), 1992. Charaka Samhita, Text With English Translation vol. II. Chaukhanba Orientalia, Varanasi.
Shen, T., Li, G.H., Wang, X.N., Lou, H.X., 2012. The genus Commiphora: a review of its traditional uses, phytochemistry and pharmacology. J. Ethnopharmacol. 142, 319–330.
Shirolkar, A., Gahlaut, A., Hooda, V., Dabur, R., 2013. Phytochemical composition changes in untreated stem juice of *Tinospora cordifolia* (W) Mier during refrigerated storage. J. Pharm. Res. 7, 1–6.
Singh, R.H., 2005. Ayurvediya Nidan Chikitsa ke Siddhanta (Hindi Version). Chowkhamba Amar Bharati Prakashan, Varanasi.
Singh, T.P., Singh, O.M., Singh, H.B., 2011. *Adhatoda vasica* Nees: phytochemical and pharmacological profile. Nat. Prod. J. 1, 29–39.
Srikantamurthy, K.R., 2003. Madhava Nidanam (Roga Vinischaya) of Madhavakara. Chaukhambha Orientali, Varanasi.
Srikanthamurthy, K.R., 1968. Luminaries of Indian Medicine, Mysore.
Srikanthamurthy, K.R., 2010. Vagbhat's Ashtanga Hridayam. Chowkhamba Krishnadas Academy, Varanasi.
Tavhare, S.D., Nishteswar, K., 2014. Collection practices of medicinal plants - Vedic, ayurvedic and modern perspectives. Int. J. Pharm. Biol. Sci. Arch. 5, 54–61.
Vakil, R.J., 1955. *Rauwolfia serpentina* in the treatment of high blood pressure; a review of the literature. Circulation 12, 220–229.
Venkatesh, P., Mukherjee, P.K., Kumar, N.S., Bandyopadhyay, A., Fukui, H., Mizuguchi, H., Islam, N., 2010. Anti-allergic activity of standardized extract of *Albizia lebbeck* with reference to catechin as a phytomarkers. Immunopharmacol. Immunotoxicol. 32, 272–276.
Wagner, H., Ulrich-Merzenich, G., 2009. Synergy research: approaching a new generation of phytopharmaceuticals. Phytomedicine 16, 97–110.
Weiss, R.S., 2009. Recipes for Immortality: Medicine, Religion, and Community in South India. Oxford University Press, New York.
Williamson, E.M., 2001. Synergy and other interactions in phytomedicines. Phytomedicine 8, 401–409.
Zahara, K., Tabassum, S., Sabir, S., Arshad, M., Qureshi, R., Amjad, M.S., Chaudhari, S. K., 2014. A review of therapeutic potential of *Saussurea lappa* - an endangered plant from Himalaya. Asian PAC J. Trop. MEd. 7S1, S60–S69.
Zimmermann, G.R., Lehar, J., Keith, C.T., 2007. Multi-target therapeutics: when the whole is greater than the sum of the parts. Drug Discov. Today 12, 34–42.